

LET AUTOMATION TAKE CONTROL

Most automation tasks require the engineer to be on-site, which means a lot of travel and outdoor work. So if you are willing to fold your sleeves, hop on to a bus and go to plants at far-flung places to learn and deliver, you can make it big in automation!



■ UMA GUPTA

India is investing hugely in infrastructure, catalysing growth of industries like cement and steel. It is also going to be the biggest drug manufacturer in the next couple of years. These industrial processes require automation to keep up with efficiency, accuracy, global norms and speed. Automation improves the safety and reliability of an industrial production process while maintaining a high product quality. With the increasing automation of industrial processes, there is need for efficient manpower as well.

Scope of automation

Anup Wadhwa, director, Automation Industry Association (AIA), explains, "Automation itself is a very vast field. Depending on the needs of businesses,

different companies classify it via different criteria. However, for industrial purposes, we talk about manufacturing environments. There are front and back ends to automation, and companies deal in the same from the end user's point of view. A company can be called to be dealing with process automation if the end user is operating in an explosive or hazardous area. Alternatively, it's called factory automation if the field and applications are related more to the general manufacturing environment."

"Automation industry addresses the design, development, production, and application of devices and systems that sense, measure and control industrial processes and manufacturing operations. This includes domains of basic continuous control, discrete, sequencing and manufacturing control, advanced control, reliability, safety and electrical controls, related software

development and integration, deployment and maintenance, he adds."

Leading academia and industry stalwarts emphasise that understanding the principles of automation will enable engineers across various industries to demonstrate new competencies that global leadership demands. These include asset management, production engineering and simulation, energy conservation, material handling and packaging, mechatronics, material and human safety, quality of work and environment, and tracking and tracing systems.

"Operator and machine safety are important issues and there are global standards and risk management practices that machine and equipment designers must begin adhering to. Industrial automation helps in implementing and monitoring safety and machine failure. It's only logical that the more critical the situation, more the care to be taken while specifying the standard or practice that needs to apply," informs Wadhwa.

Big demand, low supply

Automation industry needs people having electrical, electronics and instrumentation background. But there is a huge demand-supply gap for these engineers, though India has done a great job in terms of creating a good number of technical institutes.

"There were entrepreneurs and thinkers who realised that there is going to be a huge requirement of engineering talent in times to come. So they created a lot of private institutes and colleges, which churn out hundreds